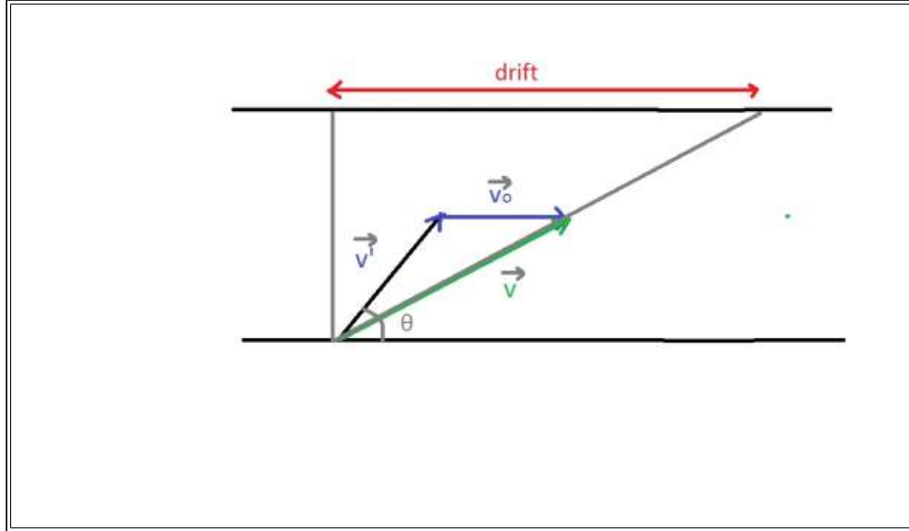


Question 1.9 : A boat moves relative to water with a velocity which is $n = 2.0$ times less than the river flow velocity. At what angle to the stream direction must the boat move to minimize drifting?

Solution: Let the river flow velocity $= v_o$. and velocity of boat relative to water would be then $v' = v_o/n$.



Across component (along y axis) of v is $v' \sin \theta$

Drift component (along x axis) of v is $v' \cos \theta + v_o$

$$\text{drift} = \frac{AB}{v' \sin \theta} (v' \cos \theta + v_o) = AB \left(\cot \theta + \frac{v_o}{v'} \operatorname{cosec} \theta \right) = AB (\cot \theta + n \operatorname{cosec} \theta)$$

Now we have to minimize drift,

$$\text{At minima, } \frac{d}{d\theta} \text{drift} = 0$$

$$\Rightarrow \frac{d}{d\theta} AB (\cot \theta + n \operatorname{cosec} \theta) = 0$$

$$\Rightarrow AB (\operatorname{cosec}^2 \theta + n \operatorname{cosec} \theta \cot \theta) = 0$$

$$\Rightarrow \cos \theta = -\frac{1}{n}$$

The only solution in $(0, \pi)$ is 120° .